Marmara University, Engineering Faculty
Electrical and Electronics Engineering
EE4078 Fundamentals of Computer Networks
First Midterm Exam Solutions, FALL2024

1.

a. When circuit switching is used, at most 8 users can be supported. This is because each circuit-switched user must be allocated its 25 Mbps bandwidth, and there is 200 Mbps of link capacity that can be allocated. (5)

b. No. Under circuit switching, the 15 users would each need to be allocated 25 Mbps, for an aggregate of 375 Mbps - more than the 200 Mbps of link capacity available. (5)

c. The probability that any 9 users of the total 15 users are transmitted and the other 6 users are idle is $\binom{15}{9}p^9(1-p)^6 = \binom{15}{9} \times 0.1^9 \times 0.9^6 = 0.0000027$ (5) (10)

2.

a. The elapsed time is: $RTT_0 + RTT_1 + RTT_2 + 2*RTT_{HTTP} = 3 + 42 + 6 + 2*83 = 217$ msecs because 2 RTT_{HTTP} are needed to fetch the HTML object - one RTT_{HTTP} to establish the TCP connection, and then one RTT_{HTTP} to perform the HTTP GET/response over that TCP connection. (5)

b. This time the elapsed time becomes $RTT_0 + RTT_1 + RTT_2 + 2*RTT_{HTTP} + 2*2*RTT_{HTTP} = 3 + 42 + 6 + 2*83 + 2*2*83 = 549$ msecs. Note that two RTT_{HTTP} delays are needed to fetch the base HTML object - one RTT_{HTTP} to establish the TCP connection, and one RTT_{HTTP} to send the HTTP request, and receive the HTTP reply. Then, serially, for each of the 2 embedded objects, a delay of $2*RTT_{HTTP}$ is needed - one RTT_{HTTP} to establish the TCP connection and then one RTT_{HTTP} to perform the HTTP GET/response over that TCP connection. (10)

c. Since there are only 2 objects, there's a delay of 51 msec for the DNS query, two RTT_{HTTP} for the base page, and 1 RTT_{HTTP} for the objects. The total is 51 + 166 + 83 = 300 msec. Two RTT_{HTTP} delays are needed to fetch the base HTML object - one RTT_{HTTP} to establish the TCP connection, and one RTT_{HTTP} to send the HTTP request, and receive the HTTP reply containing the base HTML object. However, with persistent HTTP, this TCP connection will remain open for future HTTP requests, which will therefore not incur a TCP establishment delay. Once the base object is received at the client, the maximum of five requests can proceed in parallel, each retrieving one of the 2 embedded objects. Each (in parallel) requires only one RTT_{HTTP}delay to perform the HTTP GET/response for an embedded object. Once these first five objects have been retrieved, (if necessary) the remaining embedded objects can be retrieved (in parallel). This second round of HTTP GET/response to retrieve the remaining embedded objects takes only one more RTT_{HTTP} since the TCP connection has remained open. (10)

3. (2) (2) (2) (2) (2)

a. The sender's sequence numbers are: 311, 1104, 1897, 2690, 3483

b. The receiver's ACKs are: 1104, 1897, x, x, ~~1897~~ X

(3) (3) (3) (3) (3)

4.

a. The source port number is 5984 and the destination port number is 6093. (15)

b. The source port number is 6093 and the destination port number is 6684. (10)